

CS & IT ENGINEERING

Control Flow Statements

Programming in C

Decision control statement -1

Lecture:01



By- Pankaj Sharma sir

A whiteboard with a blue border and an orange base, containing the text 'TOPICS TO BE COVERED'.

TOPICS TO BE
COVERED

A dotted orange arrow pointing from the whiteboard to the text box.

1:Decision control statement -1

int a=~~0~~¹, b=2, c=0, d=3, e;

e = ++a || (b-- && c++) && --d;

(++a) || x
↑ ↑

① e = 1 || (b-- && c++) && --d

Eval. X

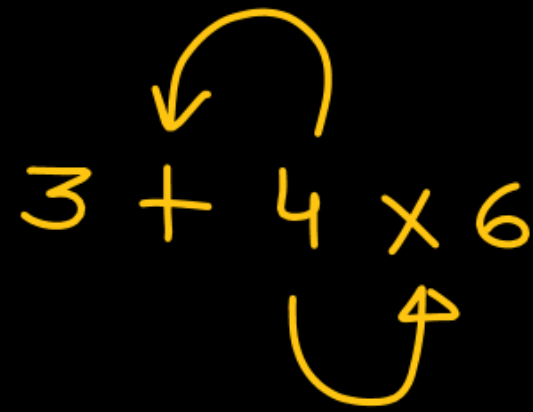
1 2 0 3 1

--, ++

++2
2 = 2 + 1 X
lvalue

(--c) --;
Error

3 + 4 x 6



3 + (4 x 6)

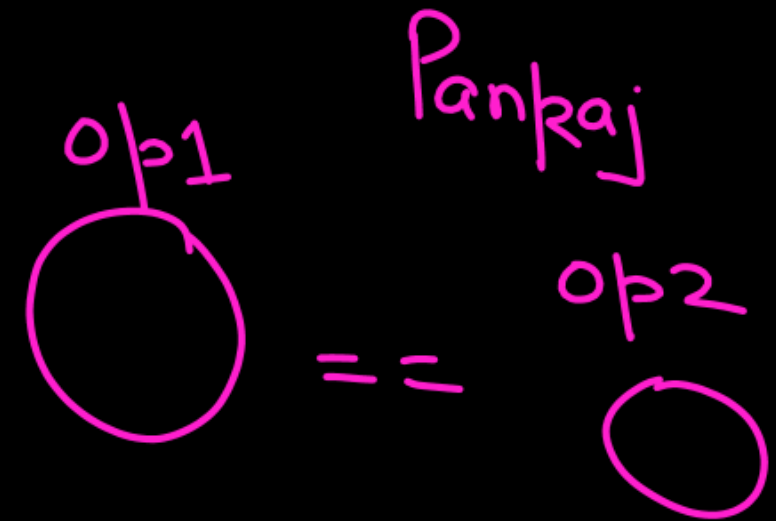
printf("/d", printf("Pankaj"))
== printf("/d", 1)

++a || x

op1 op2

Pankaj

op1 op2



Control flow statement

```
void main(){
```

```
    S1;
```

```
    S2;
```

```
    S3;
```

```
    S4;
```

```
    S5;
```

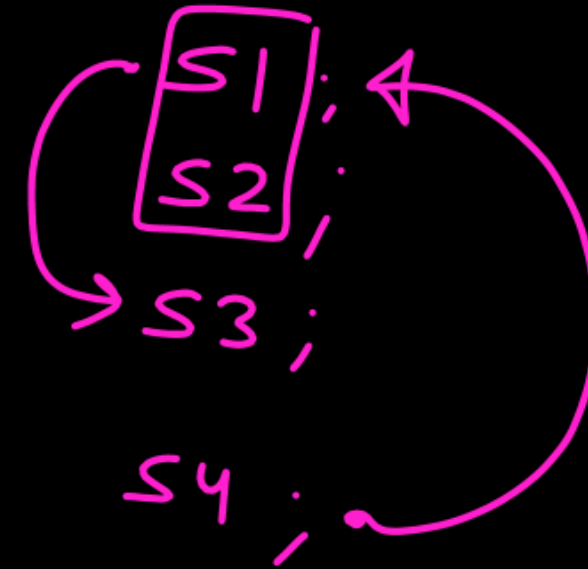
```
    S6;
```

```
    S7;
```

```
    S8;
```

```
}
```

by default
⇒ sequential



Control Flow Statement

Selection statement

if

if - else

if - else if

=

switch statement

Iterative statement
(Repetition)

- for loop

- while loop

- do while

Jump statement

break

continue

[return]
[exit]^x

Selection statement

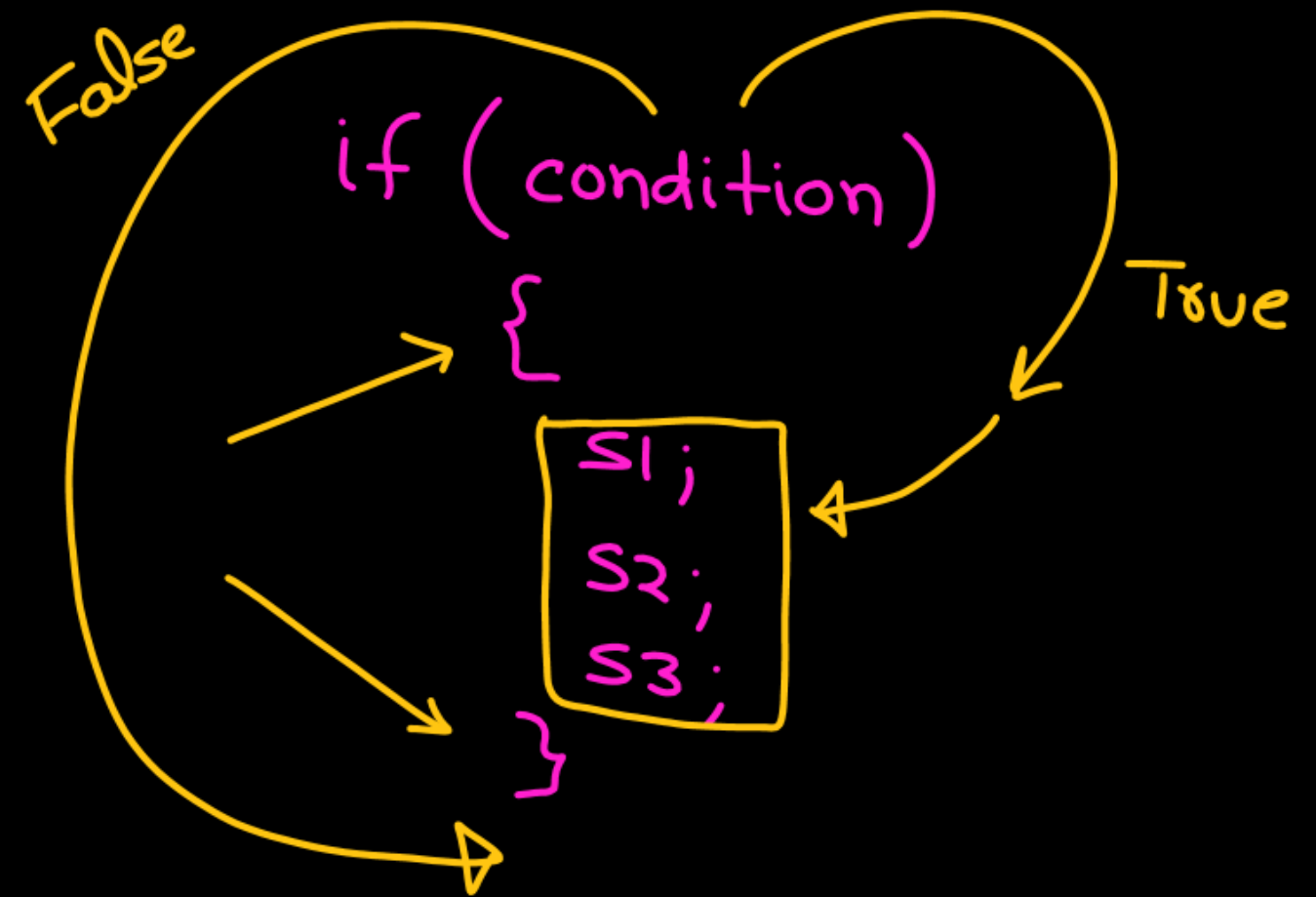
```
void main(){
```

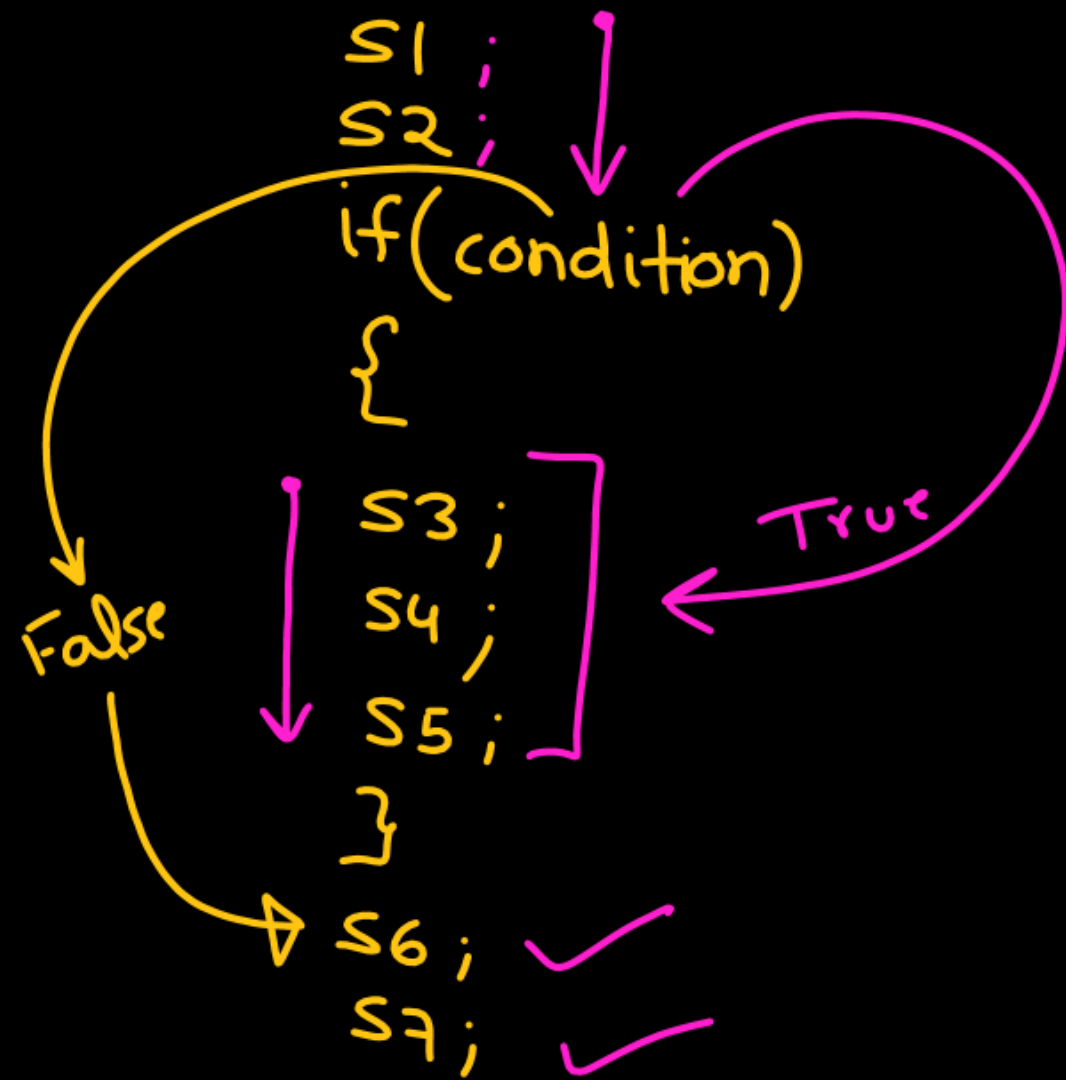
```
  ① printf("Pankaj"); ✓
```

```
  ② printf("Sharma"); ✓  
}
```

O/P: PankajSharma

if statement





$S1 \rightarrow S2 \rightarrow S3 \rightarrow S4 \rightarrow S5 \rightarrow S6 \rightarrow S7$
 $S1 \rightarrow S2 \rightarrow S6 \rightarrow S7$

Syntax :

```
if(condition/expression)
{
    Code
}
```

if(condition/exp)
S1;

Scope : First
Semi-colon


```
void main(){
```

```
    int a=10, b=12;
```

```
    printf("0"); ✓
```

```
    if(a > b){
```

```
        printf("1");
```

```
        printf("2");
```

```
    }
```

```
    printf("3"); ✓
```

```
}
```

False

10 > 12

~~True~~

False

03

```
void main(){
```

```
    int a=10, b=12;
```

```
    printf("0");
```

```
    if(a > b)
```

```
        printf("1");
```

```
        printf("2");
```

```
    printf("3");
```

```
}
```

for

⇒

Compiler

```
    ✓ printf("0");
```

```
    if(aF > b){
```

```
        printf("1"); X
```

```
    }
```

```
    ✓ printf("2");
```

```
    ✓ printf("3");
```

023

1.

```
if (2 != 3)
    printf("1");
    printf("2");
```

1 2

2.

```
✓ printf("1");
if (2) { → non-zero/True
    printf("Maza aa");
    printf("Raha");
```

1 Maza aa Raha

3.

```
printf("1");
if (! 1printf("2"))
{
    ✗ printf("3");
}
printf("4");
```

1 2 4

4 void main(){
if (printf("Pankaj"))⁶
{

}
}

Pankaj

5: int a=5;
if(a=0) → False
{
printf("O GOD");
}
printf("Please save us");

a
50

if(0) → False

a = 0
↑
Please save us

1. if(2)
{
=
}

2. if(12.78)
{
=
}

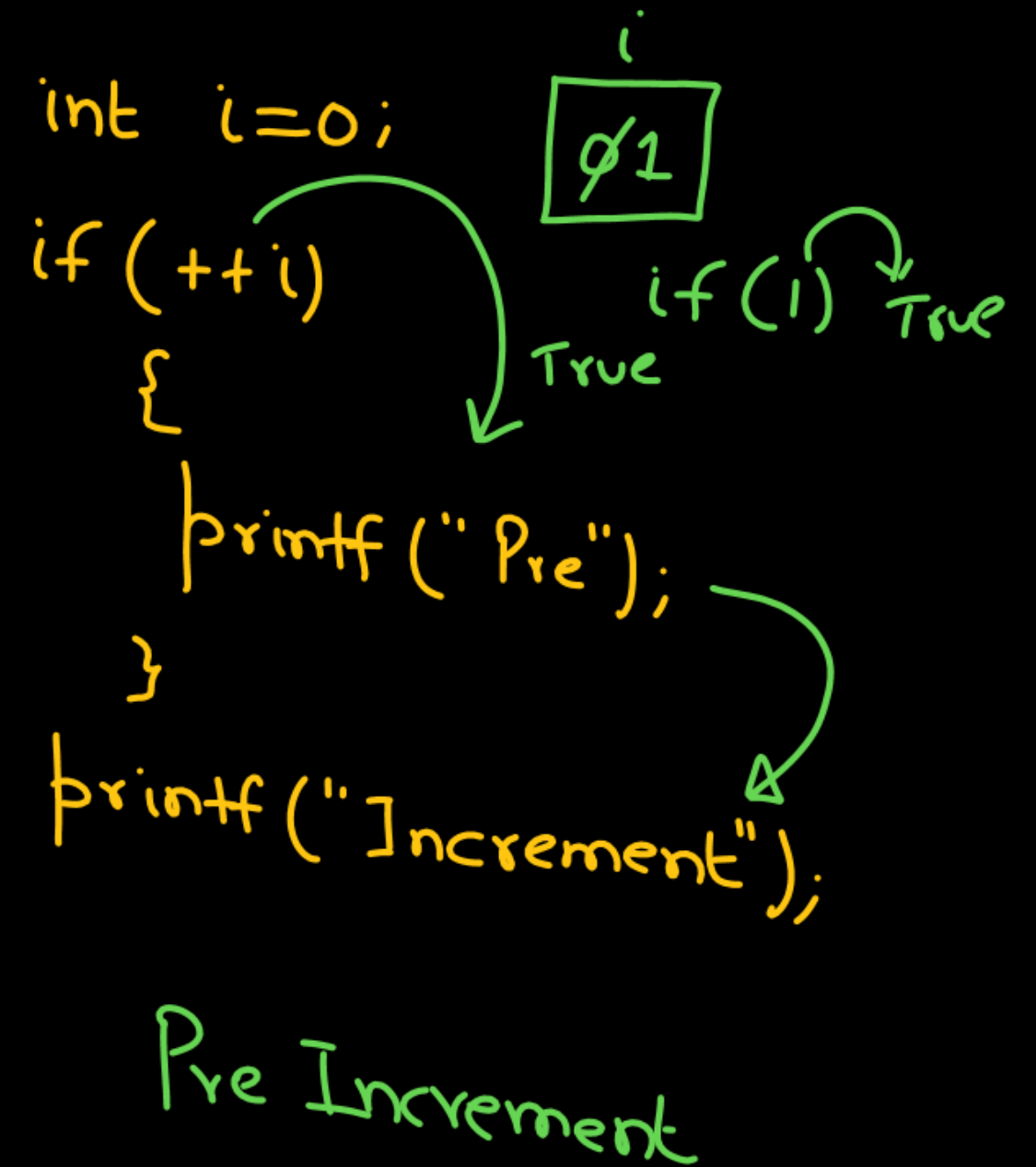
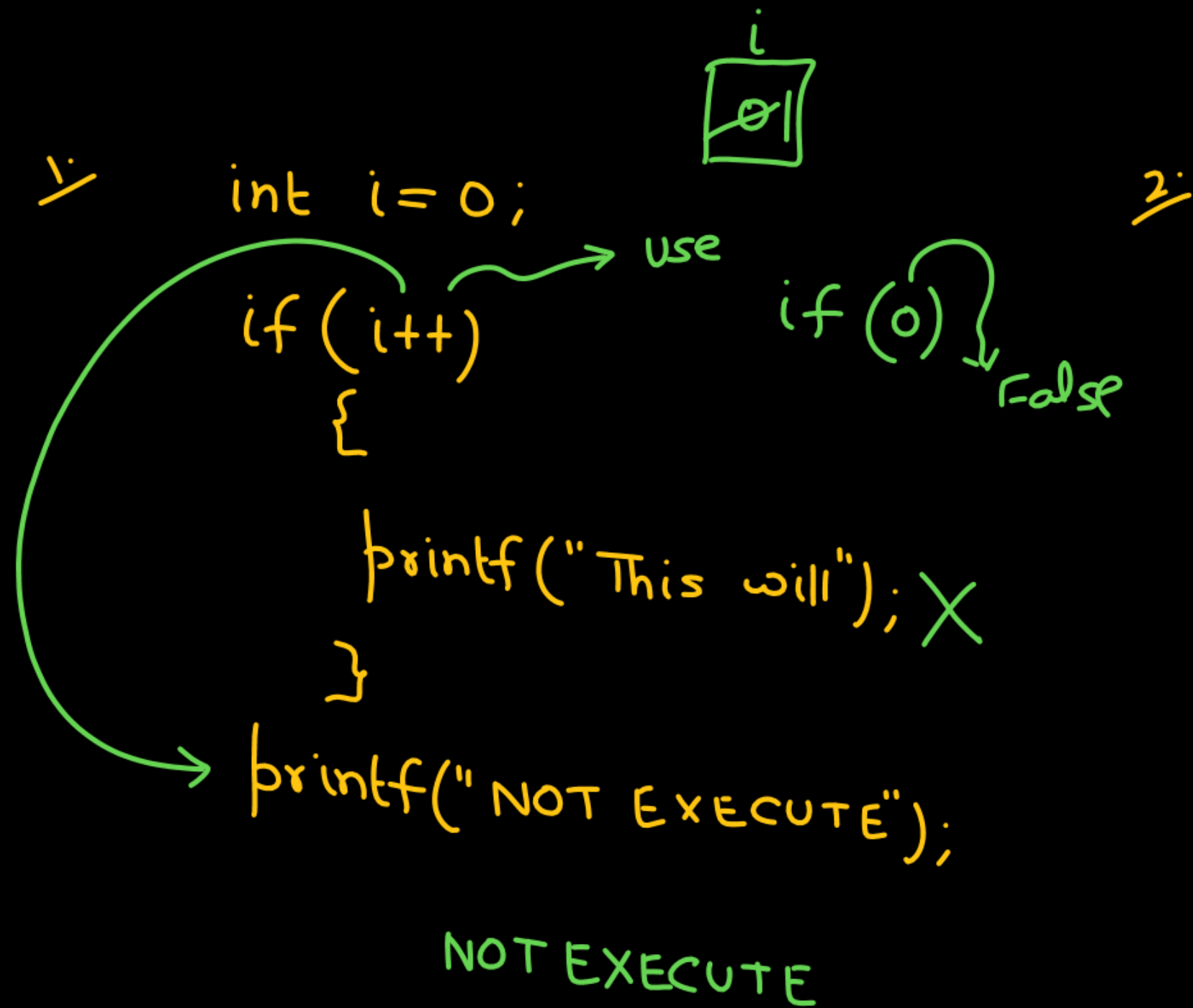
3. if(3-2+1-3.4)
{
=
}

4. if(!printf("1")+1)
{
=
}

5. int i=3;
printf("1");
if(i+3){
 printf("2");
printf("3");}

6. if()
{
=
}

→ Error



3.

int i=10, j=20, k=3;

if (i+j > k ? 10 > 6 ? 15 : 3 : 10)

{
printf("Ternary");
}

✓ printf("Operator");

if(0) {
} X

15
⇒ 0

10+20 > 3
True

(i+j > k) ? (10 > 6 ? 15 : 3) : 10
← exp1 ← exp2 ← exp3

10 > 6 ? 15 : 3
exp1 exp2 exp3
True

```
void main(){
```

```
    printf("Pankaj");
```

```
    if(0) ;
```

```
    printf("Sharma");
```

```
    }
```

O/p :

for
Compiler

```
printf("Pankaj"); ✓
```

```
if(0) {
```

```
    ;
```

```
}
```

```
printf("Sharma"); ✓
```

int a;

$a = 9 \mid = 11 > 40 ? \mid 3 \mid = 3 ? 2 > 2 \mid = 0 ? 1 : 2 : 3 : 4$
exp1 exp2 exp3

$9 \mid = 11 > 40$

$9 \mid = 0$

True

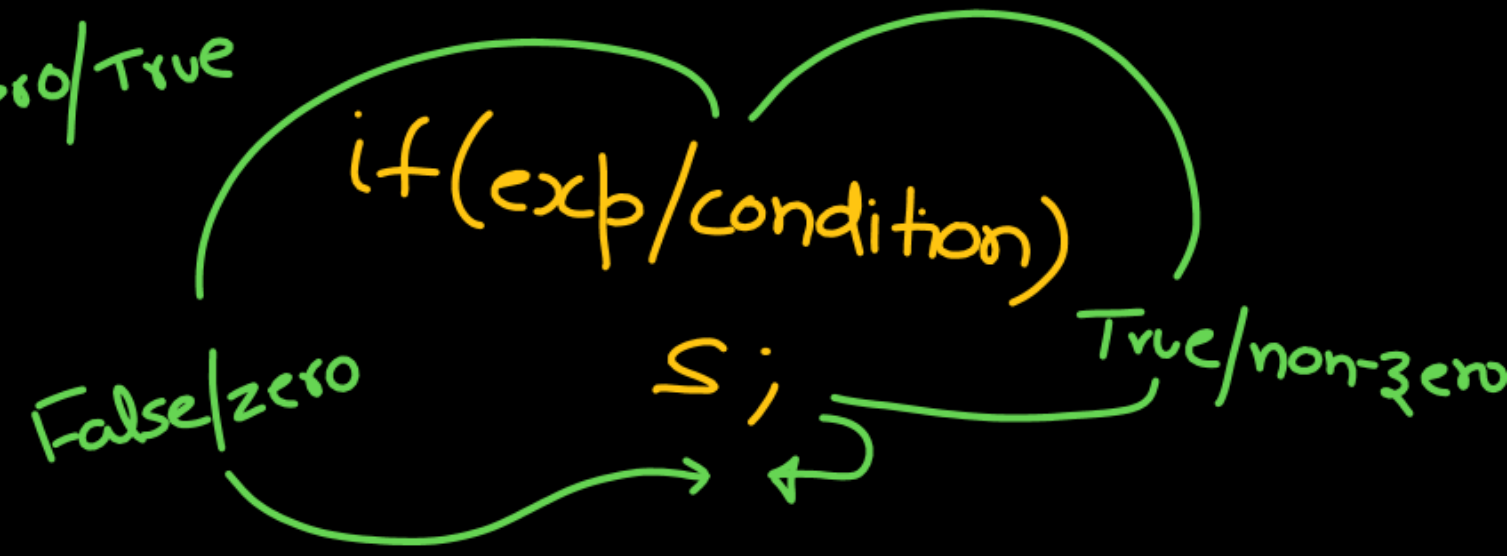
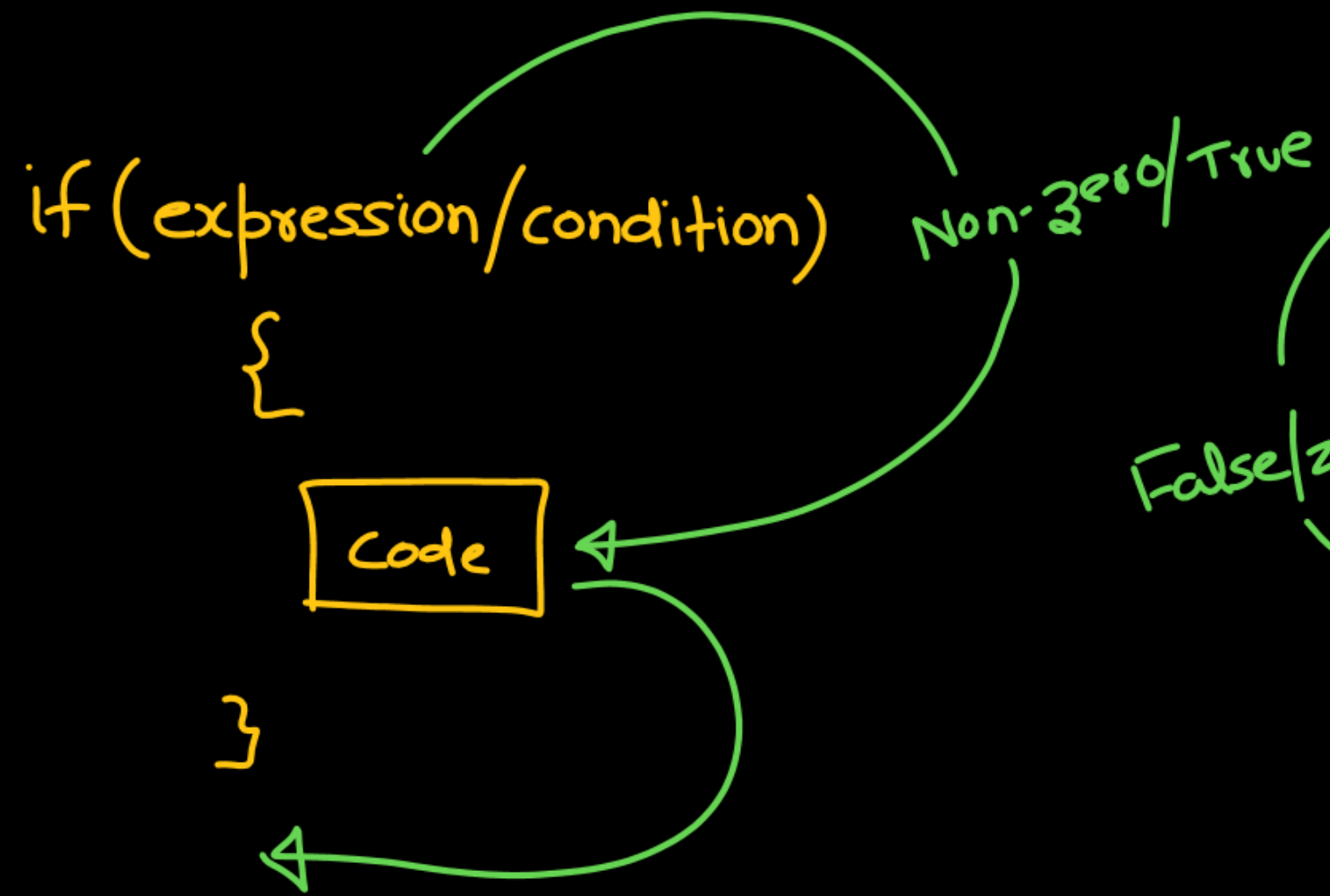
$2 > 2 \mid = 0$

$0 \mid = 0$

False

$a = \frac{1 \mid 3 \mid = 3 ? 2 > 2 \mid = 0 ? 1 : 2 : 3}{\text{exp1} \quad \text{exp2} \quad \text{exp3}}$
Unary
 $0 \mid = 3$
True

$a = \frac{2 > 2 \mid = 0 ? 1 : 2}{\text{False}} \rightarrow 2$ $a = 2$



WAP that will take a no. & if the no. is even, it will print "EVEN".

i/p : 2
o/p : EVEN

i/p : 20
o/p : EVEN

i/p : 13
o/p :

int n;
printf("Enter a number");
scanf("%d", &n);

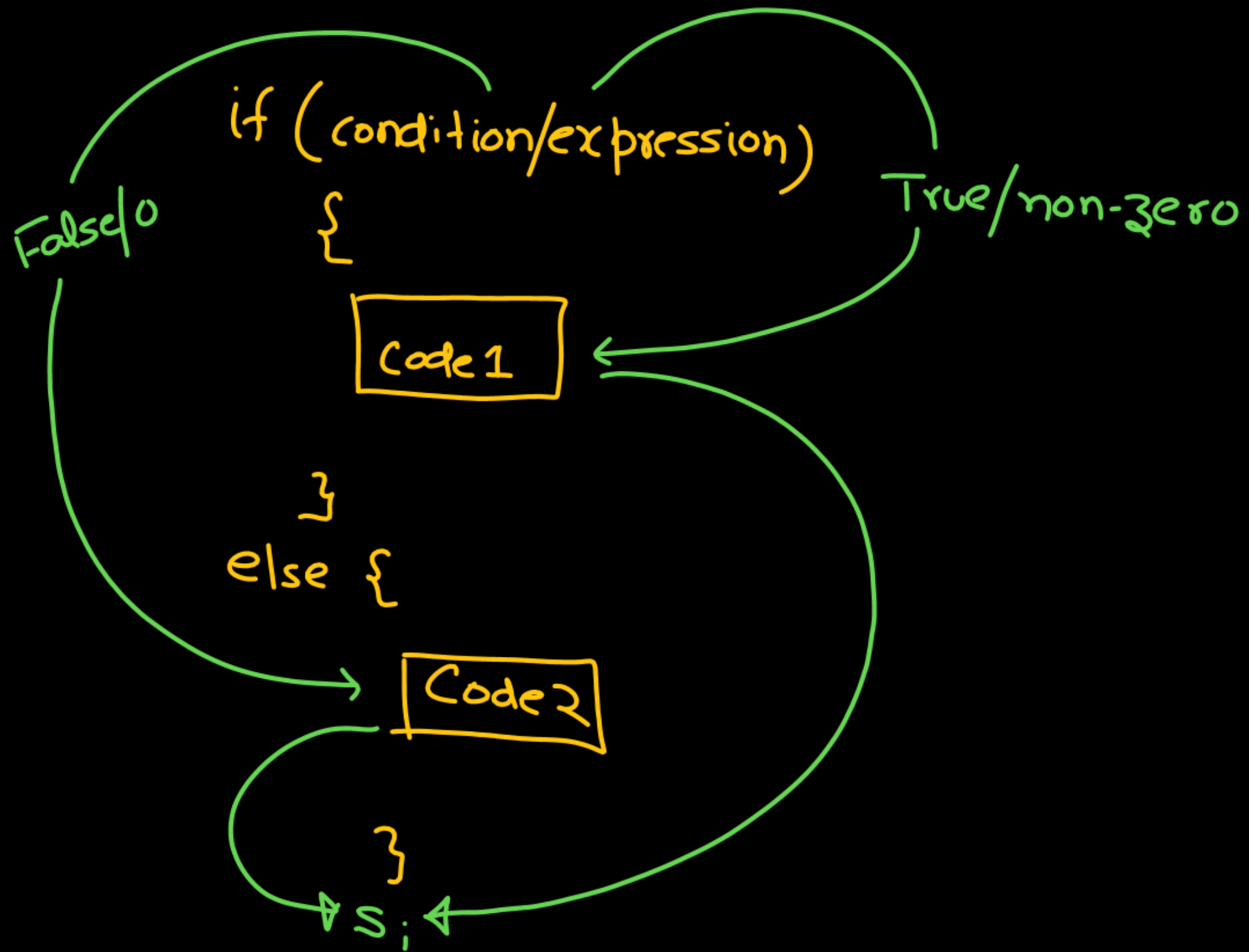
if ((n % 2) == 0)

printf("EVEN");

rem is 0
when
divided by 2

if - else

we associate an
alternate block of
code to be executed
in-case the if-condition
is false.



✓
if (! ^{! 1} printf("1"))
printf("2"); X
else
printf("3"); ✓
printf("4"); ✓

void main() {
else {
pf("Hello");
}
}

134

2.

if (2 != 3)

{
printf("Hello");
printf(" Bye");
}
else
printf("1");
printf("2");

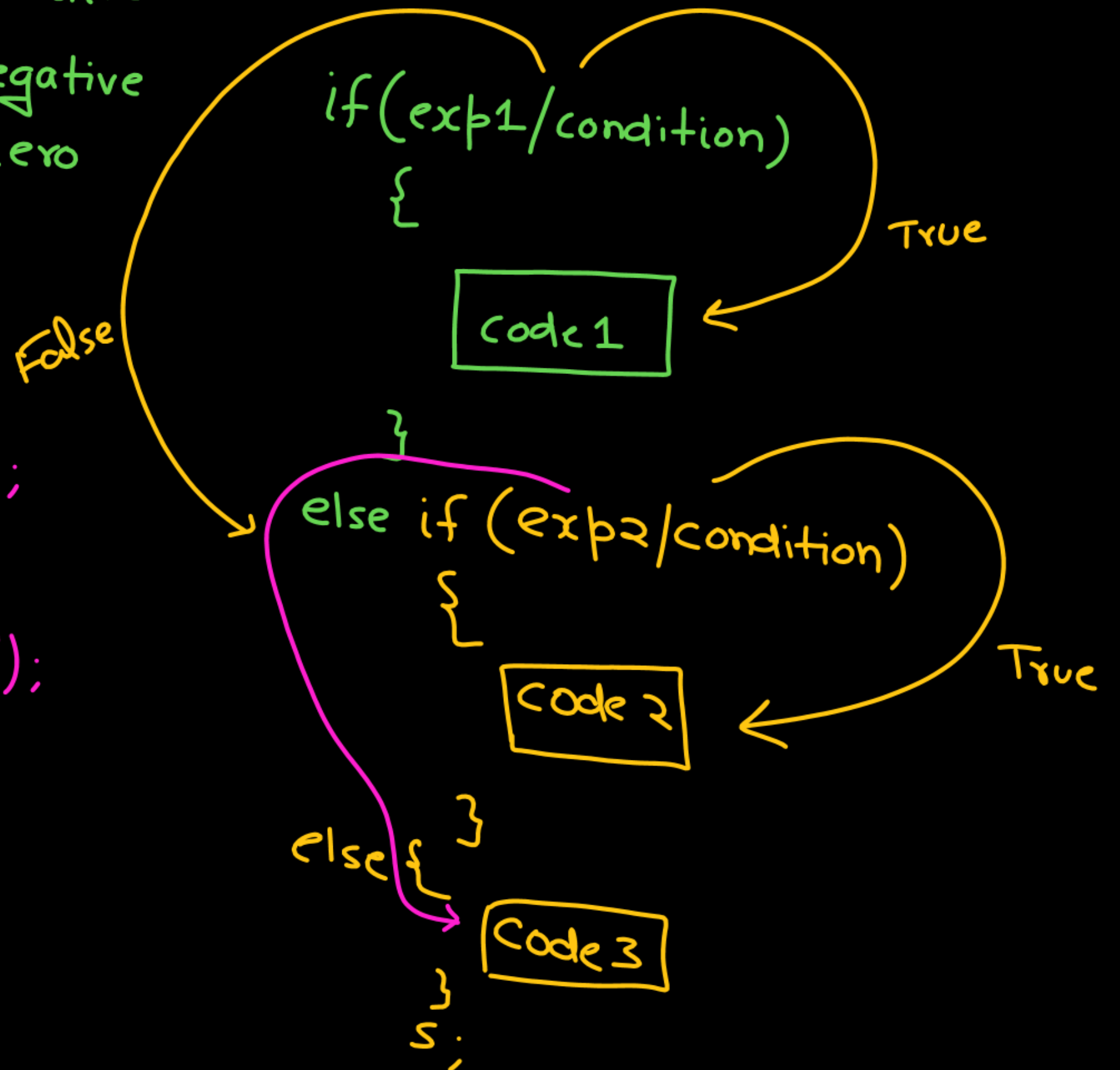
Error

WAP to take a no. from user and print whether it is odd or EVEN.

```
int n;  
printf("Enter a no");  
scanf("%d", &n);  
if (n % 2 == 0)  
    printf("%d is Even", n);  
else  
    printf("%d is Odd", n);
```

Take a no $\begin{cases} \rightarrow > 0 \Rightarrow \text{positive} \\ \rightarrow < 0 \Rightarrow \text{Negative} \\ \rightarrow = 0 \Rightarrow \text{Zero} \end{cases}$

```
if(a > 0)
    printf("Positive");
else if(a < 0)
    printf("Negative");
else
    printf("zero");
```



if(1) ✓
 printf("1"); ✓

if(2)
 printf("2"); ✓

if(3)
 printf("3"); ✓

123

if(1)
 printf("1"); ✓
else if(2)
 printf("2");
else if(3)
 printf("3");



```
int max, a = 10, b = 20;
```

```
if (a > b)
```

```
    max = a;
```

```
else
```

```
    max = b
```

```
max = (a > b) ? a : b;
```


distinct no. a, b, c

$30 > 20$ $\uparrow$ $30 > 10$
if (a > b && a > c)
 printf("/.d is largest", a); ✓
if (b > a && b > c)
 printf("/.d is largest", b);
if (c > a && c > b)
 printf("/.d is largest", c);

a = 30, b = 20, c = 10

if (a > b && a > c)
 printf("/.d is largest", a);
else {
 if (b > c)
 printf("/.d is largest", b);
 else
 printf("/.d is largest", c);
}

b or c
is
largest

distinct
 a, b, c

$$\text{largest} = (a > b \ \&\& \ a > c) ? a : (b > c) ? b : c ;$$

distinct a, b, c, d

$$\begin{aligned} \text{largest} = & (a > b \ \&\& \ a > c \ \&\& \ a > d) ? a : (b > c \ \&\& \ b > d) ? b : \\ & (c > d) ? c : d ; \end{aligned}$$

```
if (ex) {  
    =  
    =  
}
```

```
if (exp) {  
    =  
    =  
}  
else {  
    =  
    =  
}
```

```
if (exp1)  
    {  
        =  
        =  
    }  
else if (exp2)  
    {  
        =  
        =  
    }  
else if (exp3)  
    {  
        =  
        =  
    }
```

```
if (exp1)  
    {  
        =  
    }  
else if (exp2)  
    {  
        =  
        =  
    }  
else if (exp3)  
    {  
        =  
        =  
    }  
else {  
    =  
    =  
}
```

